Multiple Choice Questions (MCQs)

Total Questions: 52

I. 1. BIOLOGICAL CLASSIFICATION (Q 1 - 10)

1. Two kingdom classification was given by
    1. Whittaker 2) Linnaeus 3) Aristotle 4) Theophrastus
2. Bacteria that live in most harsh habitats are
    1. Bacteria 2) cyanobacteria 3) mycoplasmas 4) archaebacteria
3. Nitrogen Fixing Cyanobacterium is
    1. Rhizobium 2) Nostoc 3) Chlorella 4) Methanogens
4. Smallest living moneran cells that lack a cell wall are
    1. Cyanobacteria 2) protozoans 3) mycoplasma 4) bacteria
5. Diatomaceous Earth is indestructible due to cell walls embeded by
    1. Calcium 2) Silica 3) Zinc 4) Phosphorus
6. Protists that form plasmodium are
    1. Euglenoids 2) slime moulds 3) dinoflagellates 4) diatoms
7. Which is the correct sequence of the sexual cycle of fungi?
    1. Mitosis-Meiosis-Fertilisation
    2. Plasmogamy-Karyogamy-Meiosis
    3. Meiosis-Plasmogamy-Karyogamy
    4. Karyogamy-Plasmogamy-Meiosis
8. The viruses which infect bacteria are known as
    1. Zoophages 2) Bacteriophages 3) Cyanophages 4) Zymophages
9. Identify the fungi that is extensively used in biochemical and genetic work.
    1. Neurospora 2) Ustilago 3) Colletotricum 4) Saccharomyces
10. Which of the following are very good Pollution Indicators?
    1. Fungi 2) Mycoplasma 3) Lichens 4) Golden algae

II. 6. BIOMOLECULES (Q 11 - 21)

11. Chemical analysis of living organisms can be done by the usage of following chemical.
    1. Ethanol 2) Benzene 3) Trichloro acetic acid 4) Acetic acid
12. Identify the polysaccharide which is polymer of fructose.
    1. Starch 2) Glycogen 3) Cellulose 4) Inulin
13. Identify the aromatic amino acid.
    1. Glutamic acid 2) Tyrosine 3) Lysine 4) Alanine
14. Palmitic acid contains how many carbons?
    1. 16 2) 20 3) 15 4) 19
15. Cytidylic acid is a
    1. Nitrogen base 2) nucleotide 3) nucleoside 4) nucleic acid
16. Concannavalin A is a
    1. drug 2) lectin 3) alkaloid 4) toxin
17. Identify the secondary metabolite

1. Amino acid 2) Nucleic acids 3) Carbohydrates 4) Rubber

18. Which among the following is not a pyrimidine?
    1. Thymine 2) Cytosine 3) Adenine 4) Uracil
19. The following protein enables glucose uptake into cells.
    1. GLUT-4 2) Collagen 3) Antibody 4) Trypsin
20. The following structure is necessary for the many biological activities of proteins.
    1. Primary structure 2) Secondary structure 3) Tertiary structure 4) Quarternary structure
21. Which among the following is not a polymer?
    1. Polysaccharide 2) Protein 3) Nucleic acid 4) Lipid

III. 7. CELL CYCLE AND CELL DIVISION (Q 22 - 33)

22. Phase between two successive M-phases
    1. prophase 2) metaphase 3) inter phase 4) anaphase
23. Cell cycle duration of yeast
    1. 24 hours 2) 90 minutes 3) 20 minutes 4) 60 minutes
24. During 'S' phase of interphase
    1. DNA synthesizes, chromosome number remains same
    2. DNA synthesizes, chromosome number doubled
    3. DNA does not synthesize, chromosome number reduced to half
    4. DNA does not synthesize, chromosome number remains same
25. A cell in G1 phase has 16 chromosomes, how many chromosomes does the cell have in G2 phase.
    1. 24 2) 32 3) 8 4) 16
26. The cells in quiescent stage (G0 phase) exit the cell cycle from the following phase.
    1. G2 phase 2) G1 phase 3) S phase 4) M phase
27. Identify the correct sequential phases in the Karyokinesis.
    1. Prophase, metaphase, anaphase, telophase
    2. Metaphase, anaphase, telophase, prophase
    3. Anaphase, telophase, prophase, metaphase
    4. Telophase, prophase, metaphase, anaphase
28. Centromere splitting occurring in
    1. mitotic anaphase and meiotic anaphase-I
    2. mitotic anaphase and meiotic anaphase-II
    3. meiotic anaphase I and meiotic anaphase-II
    4. mitotic anaphase only
29. Compaction of chromosomes start from
    1. zygotene 2) pachytene 3) leptotene 4) diplotene
30. Bivalent formation occurs in the following stage
    1. zygotene 2) pachytene 3) leptotene 4) diplotene
31. The X-shaped structures called chiasmata appears in
    1. zygotene 2) pachytene 3) leptotene 4) diplotene
32. Terminalisation of chiasmata happens in
    1. zygotene 2) pachytene 3) diplotene 4) diakinesis
33. A bivalent in meiosis-I consists of

1. one chromosome, two chromatids
2. two chromosomes, two chromatids
3. two chromosomes, four chromatids
4. four chromosomes, four chromatids

IV. 8. PHOTOSYNTHESIS IN HIGHER PLANTS (Q 34 - 42)

34. Who discovered oxygen?
    1. Joseph Priestly 2) T. Engelmann 3) Jan Ingenhousz 4) Van Neil
35. Chloroplast is
    1. single membrane bound organelle
    2. double membrane bound organelle
    3. triple membrane bound organelle
    4. membrane lacking organelle
36. Which is the most abundant plant pigment in the world?
    1. Chlorophyll a 2) Chlorophyll b 3) Carotenoids 4) Xanthophylls
37. Maximum absorption by chlorophyll a occurs in
    1. blue & green region
    2. red & green region
    3. blue & red region
    4. yellow & red region
38. LHC stands for
    1. Late Harvesting Complex
    2. Light Harvesting Complex
    3. Light Hanging Complex
    4. Late Hanging Complex
39. During photosynthesis the O2 is released in
    1. lumen of thylakoid 2) outer side of thylakoid 3) stroma 4) cytoplasm
40. For formation of 1 glucose molecule, how many turns of Calvin cycle is/are needed?
    1. 3 2) 1 3) 2 4) 6
41. The most crucial step of Calvin cycle is
    1. carbonation 2) carboxylation 3) reduction 4) regeneration
42. The first stable product during CO2 fixation in C4 cycle is
    1. RBP 2) PEP 3) OAA 4) PGA

V. 9. RESPIRATION IN PLANTS (Q 43 - 52)

43. Glycolysis is also known as _________pathway.
    1. ETS 2) EMP 3) ENP 4) ELP
44. End product of glycolysis is
    1. pyruvic acid 2) oxalo acetic acid 3) phosphoenolpyruvic acid 4) citric acid
45. Fermentation occurs when there is
    1. complete supply of oxygen
    2. no supply of oxygen
    3. inadequate supply of oxygen
    4. complete supply of water

46. In alcoholic fermentation, pyruvate is converted to which among the following?
    1. Ethanol, CO_2, NADH
    2. CO_2 and Ethanol
    3. CO_2 and Ethanol only
    4. Ethanol, CO_2, $NADH_2$
47. Which enzyme catalyses the oxidative decarboxylation of pyruvic acid?
    1. Pyruvate carboxylase
    2. Lactate dehydrogenase
    3. Alcohol dehydrogenase
    4. Pyruvate dehydrogenase
48. Where does TCA cycle occur?
    1. Cytoplasm
    2. Inner membrane of mitochondria
    3. Mitochondrial matrix
    4. Stroma of chloroplast
49. What is the first formed compound in TCA cycle?
    1. Acetyl CoA 2) Citric acid 3) Isocitric acid 4) OAA
50. Which among the following is synthesized during the conversion of succinyl - CoA to succinic acid in TCA cycle?
    1. FADH 2) GTP 3) $NADH_2$ 4) $NADPH_2$
51. How many total $NADH_2$ are produced from a pyruvic acid after completion of TCA cycle?
    1. 2 2) 3 3) 4 4) 5
52. The site of ETC is
    1. cytoplasm
    2. inner membrane of mitochondria
    3. outer membrane of mitochondria
    4. matrix of mitochondria

II. Fill in the Blanks (FIB)

Total Questions: 50

I. 1. BIOLOGICAL CLASSIFICATION (FIB 1 - 10)

1. In five kingdom classification, Bacteria are included in ___________kingdom.
2. Methanogens are present in the gut of several ruminant animals such as cows and buffaloes and they are responsible for the production of ___________.
3. The colonies of cyanobacteria are generally surrounded by ___________sheath.
4. Bacteria reproduce mainly by ___________in the oceans.
5. Diatoms are the chief ___________organisms.
6. Pellicle is found in ___________.
7. The network of hyphae is known as ___________.
8. Agaricus belongs to ___________fungi.
9. ___________fungi is known as imperfect fungi.
10. Viroids were discovered by ___________.

II. 6. BIOMOLECULES (FIB 11 - 20)

11. The pentose sugar in DNA is __________.
12. Nucleic acids percentage in the total cellular mass is __________.
13. Most abundant protein in the biosphere __________.
14. The molecular weight of macromolecules is greater than __________Daltons.
15. The amino acids in protein are linked by __________bond.
16. Exo skeleton of arthropods is made up of __________.
17. The R-group of the serine is __________.
18. The micro molecule which is found in acid insoluble fraction __________.
19. Coenzyme NAD and NADP contain __________vitamin.
20. Most of the enzymes get damaged/denatured above __________degrees temperature.

III. 7. CELL CYCLE AND CELL DIVISION (FIB 21 - 30)

21. Most suitable phase of cell division to study the morphology of chromosome is __________.
22. Prophase-I of meiosis has leptotene __________pachytene, diplotene and diakinesis.
23. The stage between meiosis -I and meiosis-II is __________.
24. Crossing over happens in the __________of prophase - I.
25. In meiosis splitting of centromere in a chromosome happens during __________.
26. What is the longest phase in the oocytes of some vertebrates __________.
27. The disc shaped structures at the surface of the centromeres are called __________.
28. Cell furrow forms during cytokinesis of __________cell.
29. The duration of M-phase in human cell is __________.
30. Which division increases the genetic variability in the population __________.

IV. 8. PHOTOSYNTHESIS IN HIGHER PLANTS (FIB 31 - 40)

31. __________in 1770 performed a series of experiments that revealed the essential role of air in the growth of green plants.
32. Chlorophyll __________is the chief pigment associated with photosynthesis.
33. In PS-I the reaction centre chlorophyll-'a' has an absorption peak at __________nm.
34. The splitting of water is associated with the PS- __________.
35. The __________enzyme is located on the stroma side of the membrane.
36. Chemiosmosis is helpful for the synthesis of __________.
37. The products of light reaction are ATP, NADPH and __________.
38. RuBP carboxylase also has an oxygenation activity, so it would be more correct to call it __________.
39. Kranz' anatomy is characteristic feature of __________plants.
40. __________is the major limiting factor for photosynthesis.

V. 9. RESPIRATION IN PLANTS (FIB 41 - 50)

41. The energy currency of the cell is __________.
42. Glycolysis occurs in __________.
43. In muscle cells of animals, when oxygen is inadequate, pyruvic acid is converted to

___________.
44. Pyruvic acid + CoA + NAD → Acetyl CoA + CO2 + NADH + H+ Above reaction is catalyzed by ___________Enzyme.
45. During the conversion of succinyl-CoA to succinic acid in citric acid cycle, a molecule of ___________is synthesized.
46. Tricarboxylic acid cycle was discovered by ___________.
47. ATP synthase is located on ___________membrane of Mitochondria.
48. Oxidation of one molecule of FADH2 give rise to ___________molecules of ATP.
49. Respiratory pathway is an ___________pathway as it involves both catabolism and anabolism.
50. The ratio of the volume of CO2 evolved to the volume of O2 consumed in respiration is called ___________.

III. One Word Answer Questions

Total Questions: 53

I. 1. BIOLOGICAL CLASSIFICATION (OWA 1 - 10)

1. Among five kingdom classification, eukaryotes are placed in how many kingdoms?
2. What is the cell wall composition of fungi?
3. Give one example of Nitrogen fixing cyanobacteria.
4. What is the protist responsible for Red tides?
5. What are the smallest living cells which can survive without oxygen?
6. What is the causative organism for Sleeping sickness disease?
7. Which protistans show both autotrophic and heterotrophic nutrition?
8. Agaricus belongs to which class of the fungi?
9. Which organism causes mad cow disease?
10. Who proposed the name "Contagium vivum fluidum" (Infectious living fluid)?

II. 6. BIOMOLECULES (OWA 11 - 20)

11. Name the pyrimidine which is absent in the RNA.
12. Name the acid which is formed in our skeletal muscle, under anaerobic conditions.
13. In the presence of carbonic anhydrase how many H_2CO_3 molecules can be formed per second.
14. What is the metal ion cofactor for the proteolytic enzyme carboxypeptidase?
15. Name the non-protein constituent of the enzyme.
16. What is the most abundant protein in the animal world?
17. Among the starch and cellulose which one holds iodine and gives blue colour?
18. Nucleic acids with catalytic power are known as?
19. Name the first and last amino acids of a polypeptide chain.
20. Enzymes are categorised into how many classes?

III. 7. CELL CYCLE AND CELL DIVISION (OWA 21 - 33)

21. Name the two phases of cell cycle.
22. What is the approximate duration of human cell cycle?
23. Which of the phases of cell cycle is of longest duration?
24. In which stage of cell cycle does DNA synthesis occur?
25. Name the stage of meiosis in which actual reduction in chromosome number occurs.
26. What is the DNA content in a cell after completion of 'S' phase of interphase, when taken 2C as in its G1 phase?
27. Centriole duplication occurs in which phase of interphase?
28. Is the cell in quiescent stage (G0) metabolically active or not?
29. What type of cell division is also known as equational division?
30. How many chromatids does the prophase chromosome have?
31. An anther has 1200 pollen grains. How many pollen mother cells must have been there to produce them?
32. Given that the average duplication time of E.coli is 20 minutes. How much time will two E. coli cells take to become 32 cells?
33. If a tissue has at a given time 1024 cells. How many cycles of mitosis had the original parental single cell undergone?

IV. 8. PHOTOSYNTHESIS IN HIGHER PLANTS (OWA 34 - 43)

34. What is the primary acceptor of CO_2 in C3 plants?
35. Which is the first formed compound in C3 plants?
36. What is the primary acceptor of CO_2 in C4 plants?
37. Name the first formed compound in C4 plants.
38. Where does photolysis of water take place in Chloroplast?
39. Who proposed law of limiting factor in photosynthesis?
40. How many ATP are required for synthesis of one glucose in C3 plants?
41. Where does light reaction takes place in chloroplast?
42. Where does dark reaction takes place in chloroplast?
43. Name the first enzyme that is involved in carboxylation process in C3 Plants.

V. 9. RESPIRATION IN PLANTS (OWA 44 - 53)

44. Which enzyme catalyses the phosphorylation of glucose during glycolysis?
45. Where is electron transport system located in mitochondria of cell?
46. Name the co-factors required for the activity of pyruvate dehydrogenase.
47. How many number of ATP produced when pyruvate is converted to lactate by fermentation?
48. In glucose, how many carbon molecules are present?
49. What is the end product of anaerobic respiration in yeast?
50. Respiratory quotient of carbohydrates is 1. why?
51. Glycolysis occurs in which part of the cell?
52. Name the process of conversion of glucose into ethyl alcohol.
53. Which compound serves as connecting link between glycolysis and Kreb's cycle?

IV. Very Short Answer Questions (VSAQs)

Total Questions: 63

I. 1. BIOLOGICAL CLASSIFICATION (VSAQ 1 - 7)

1. What is the nature of cell wall in Diatoms?
2. What do the term “phycobiont” and “mycobiont” refer” to?
3. What do the terms 'Algal blooms' and 'Red tides' signify?
4. What are 'Viroids' different from 'Viruses'?
5. State two economically important uses of Heterotrophic bacteria.
6. Some plants are autotrophic. Can you think of some plants that are partially heterotrophic?
7. Give the main criteria used for classification by Whittaker.

II. 3. MORPHOLOGY OF FLOWERING PLANTS (VSAQ 8 - 23)

8. Differentiate fibrous roots from adventitious roots.
9. What is meant by pulvinus leaf base? In members of which angiosperm family do you find them?
10. How do dicots differ from monocots with respect to venation?
11. How is a pinnately compound leaf different from a palmately compound leaf? Explain with one example each.
12. Differentiate between Racemose and Cymose Inflorescences.
13. Differentiate actinomorphic from zygomorphic flower.
14. How do the petals in pea plant are arranged? What is such type of arrangement called?
15. What is meant by epipetalous condition? Give an example.
16. Differentiate between apocarpous and syncarpous ovary.
17. Define placentation. What type of placentation is found in Dianthus?
18. What is meant by parthenocarpic fruit?
19. What is the type of fruit found in mango? How does it differ from that of coconut?
20. What is meant by scutellum? In which type of seeds is it present?
21. Define with examples endospermic and non-endospermic seeds.
22. Write the floral formula of Solanum plant.
23. Give the technical description of ovary in Solanum nigrum.

III. 5. CELL: THE UNIT OF LIFE (VSAQ 24 - 34)

24. What is the significance of vacuole in a plant cell?
25. What does 'S' refer in 70S & 80S ribosomes?
26. What is the function of a polysome?
27. What is referred to as satellite in some chromosomes?
28. What is middle lamella made of? What is its functional significance?
29. What is osmosis?

30. Name the cell organelle which is called power house of the cells.
31. Which part of the bacterial cell is targeted in gram staining?
32. Which type of ribosomes are present in chloroplast and mitochondria?
33. Name the fat-soluble pigments which are present in chromoplast.
34. Match the following:
    (a) Cristae
    (b) Mesosome
    (c) Thylakoids
    (d) Cisternae
    (i) Flat membranous sacs in stroma of chloroplast
    (ii) Infoldings in mitochondria
    (iii) Disc-shaped sacs in Golgi apparatus
    (iv) Infoldings of plasma membrane in bacteria

IV. 6. BIOMOLECULES (VSAQ 35 - 43)

35. Give one example for each of amino acids, sugars, nucleotides and fatty acids.
36. Explain the zwitterionic form of an amino acid.
37. Glycine and alanine are different with respect to one substituent on the alpha carbon. What are the other common substituent groups?
38. Starch, cellulose, glycogen, chitin are polysaccharides found among the following. Choose the one appropriate against each:
    a. Cotton fibre
    b. Exo skeleton of cockroach
    c. Liver
    d. Peeled potato
39. What are primary, secondary metabolites? Give examples.
40. Distinguish between apoenzyme and cofactor.
41. How are prosthetic groups different from co-enzyme?
42. What are competitive enzyme inhibitors? Mention one example.
43. Why are 'Oxidoreductases', so named?

V. 8. PHOTOSYNTHESIS IN HIGHER PLANTS (VSAQ 44 - 53)

44. Name the processes which take place in grana and stroma regions of chloroplast.
45. Where does the photolysis of H_2O occur? what is its significance?
46. Where is the enzyme NADP reductase located? What is released if the proton gradient breaks down?
47. Explain the terms: (a) PEP Carboxylase (b) Bundle sheath cells.
48. Mention the components of ATPase enzyme. What is their location? Which part of the enzyme shows conformational change?
49. Distinguish between action spectrum and absorption spectrum.
50. Out of the basic raw materials of photosynthesis, what is reduced? What is oxidized?
51. Define the law of limiting factors proposed by Blackman.
52. What is the primary accepter of CO_2 in C3 plants? What is first stable compound formed in Calvin cycle?

53. What is the primary accepter of CO_2 in C4 plants? What is the first compound formed as a result of primary carboxylation in the C4 pathway?

VI. 10. PLANT GROWTH AND DEVELOPMENT (VSAQ 54 - 63)

54. Define plasticity. Give an example.
55. What is the disease that formed the basis of the identification of gibberellins in plants? Name the causative fungus of this disease.
56. What is apical dominance? Name the growth hormone that causes it.
57. What is meant by bolting? Which hormone causes bolting?
58. Define respiratory climactic? Name the PGR associated with it.
59. What is ethephon? Write its role in agricultural practices.
60. Why is abscisic acid also known as stress hormone?
61. Define growth and development.
62. Name the phases observed in sigmoid growth curve.
63. What are the very essential conditions required for growth?

I. Short Answer Questions (SAQs)

Total Questions: 36

I. 2. PLANT KINGDOM (SAQ 1 - 6)

1. Differentiate between red algae and brown algae.
2. Differentiate between liverworts and mosses.
3. What is meant by homosporous and heterosporous Pteridophytes? Give two examples.
4. Give a brief account of prothallus.
5. Write a note on economic importance of Algae and Bryophytes.
6. Draw labelled diagrams of:
   a. Female thallus and male thallus of a liverwort
   b. Gametophyte and sporophyte of Funaria.

II. 3. MORPHOLOGY OF FLOWERING PLANTS (SAQ 7 - 14)

7. Explain different regions of root with neat labeled diagram.
8. Explain different types of phyllotaxy with examples.
9. Describe the arrangement of floral members in relation to their insertion on the thalamus.
10. Describe the four types of placentations found in flowering plants with respect to the arrangement of sepals and petals in respective whorls. Explain.
11. The flowers of many angiospermic plants show sepals and petals, differ with respect to the arrangement of sepals and petals in respective whorls. Explain.
12. Write a brief note on semi-technical description of a typical flowering plant.
13. Give an account of floral diagram.
14. Draw the labelled diagram of the following:
    i) Gram seed

ii) V.S. of maize seed

III. 5. CELL: THE UNIT OF LIFE (SAQ 15 - 23)

15. Briefly describe the cell theory.
16. Give the biochemical composition of plasma membrane. How are lipid molecules arranged in the membrane?
17. Distinguish between active transport and passive transport?
18. What are the characteristics of a prokaryotic cell?
19. Explain the structure of the cell organelle which contain chlorophyll, with a neat labelled diagram.
20. Explain the structure of the cell organelle which is known as 'Power house of the cell' with a neat labelled diagram.
21. Differentiate between rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER) with a neat labelled diagram.
22. Explain the structure of the nucleus with a neat labelled diagram.
23. Classify the chromosomes on the basis of centromere location.

IV. 7. CELL CYCLE AND CELL DIVISION (SAQ 24 - 27)

24. In which phase of meiosis the following are formed? Choose the answers from hint points given below.
    a. Synaptonemal complex
    b. Recombination nodules
    c. Functioning of the enzyme recombinase
    d. Terminalization of chiasmata
    e. Interkinesis
    f. Formation of dyad of cells
    Hints: i. Pachytene, ii. After Telophase-I / before Meiosis-II, iii. Zygotene, iv. Diakinesis, v. Pachytene.
25. Mitosis results in producing two daughter cells which are similar to each other. What would be the consequence if each of the following irregularities occurs during mitosis?
    a. Nuclear membrane fails to disintegrate
    b. Duplication of DNA does not occur
    c. Centromere does not divide
    d. Cytokinesis does not occur
26. Describe the events in prophase-I of meiosis.
27. Though redundantly described as a resting phase, interphase does not really involve rest. Comment.

V. 8. PHOTOSYNTHESIS IN HIGHER PLANTS (SAQ 28 - 32)

28. Differentiate between C3 and C4 plants.
29. Write short notes on photorespiration.
30. Describe the mechanism of C4 pathway.
31. Draw neat labelled diagram of chloroplast.
32. Differentiate between cyclic and non-cyclic photophosphorylation.

VI. 10. PLANT GROWTH AND DEVELOPMENT (SAQ 33 - 36)

33. Write a note on agricultural and horticultural applications of auxins.
34. Write the physiological responses of the gibberellins in plants.
35. Write any four physiological effects of cytokinins in plants.
36. What are the physiological processes that are regulated by ethylene in plants?